

SUNIL HANSDA

+91 7903074075 | sunilhansda00001@gmail.com | [GitHub](#) | [LinkedIn](#)

PROFILE SUMMARY

Backend Engineer with 3 years of experience building scalable backend systems using **Python (FastAPI, Flask)** and cloud-native technologies. Experienced in designing **RESTful APIs**, microservices, distributed systems, and high-performance database solutions for production environments. Skilled in **SQL/NoSQL databases**, CI/CD automation, GCP deployments, and integrating Generative AI capabilities using **LangGraph, LangChain, and RAG pipelines**.

SKILLS

Programming Languages: Python, Java, Shell Scripting, SQL, C++

Backend & APIs: FastAPI, Flask, REST APIs, Pytest, GraphQL, Microservices, AsyncIO, API Design, Distributed Systems, Design Patterns, Object-Oriented Programming, JWT Authentication

AI Integration & Frameworks: LangChain, LangGraph, RAG Pipelines, Agentic AI, Prompt Engineering, Vector Databases (Pinecone)

Databases, Messaging & Frontend: PostgreSQL, MySQL, MongoDB, SQL Server, Redis, Kafka, Angular

Cloud & DevOps: GCP, Azure, Docker, Kubernetes, CI/CD, Linux, Git, GitLab, HashiCorp Vault

WORK EXPERIENCE

CGI Inc., Bangalore

July 2023 – Present

Software Engineer

ETL Migration Platform

- Architected and delivered FastAPI-based backend services supporting metadata-driven ETL migration and enterprise system integrations.
- Engineered 15+ async RESTful APIs using FastAPI and AsyncIO for pipeline orchestration, execution tracking, metadata governance, and concurrent automated processing.
- Designed and deployed agentic AI workflows using LangGraph, orchestrating multi-step pipeline configuration generation across 200+ migration workflows, reducing manual setup effort by 40%.
- Established observability through structured logging, tracing, and monitoring, improving system reliability and accelerating root-cause analysis.
- Optimized database schemas and complex SQL queries across MySQL and SQL Server, reducing metadata retrieval latency by almost 30%.
- Introduced metadata-driven configuration frameworks that saved approximately 2 sprint points per release cycle and accelerated onboarding of new migration initiatives.
- Wrote unit and integration tests using pytest to validate pipeline orchestration logic, metadata transformations, and API contract behaviour across services.
- Tech Stack:* Python, FastAPI, LangGraph, LangChain, MySQL, SQL Server, Git

DevOps & Systems Reliability

- Owned backend capabilities throughout the software development lifecycle, from design and implementation to production deployment support.
- Administered GCP-hosted services across development, staging, and production environments, ensuring operational stability.
- Resolved production incidents through root-cause analysis and performance tuning of critical backend services.
- Applied Git-based engineering practices, including code reviews, branching strategies, and CI/CD workflows.
- Integrated HashiCorp Vault with GitLab CI pipelines to eliminate manual credential management and strengthen security compliance.
- Streamlined deployment workflows for 1200+ ETL artifacts using Docker and GitLab runners, reducing deployment effort by 50%.
- Supported distributed systems utilizing Kafka and Redis for asynchronous processing, caching, and reliable inter-service communication.
- Tech Stack:* GCP, Docker, Kubernetes, Redis, Kafka, Linux, CI/CD, GitLab

Internal Data Platform & Tooling

- Created internal backend tooling for synthetic data generation supporting ETL development, testing, and validation activities.
- Automated schema-aware dataset generation, reducing dependency on upstream teams and saving over 20 engineering hours per week.
- Applied LLM-based automation and metadata grounding techniques to generate context-aware datasets and simplify preprocessing tasks.
- Delivered configurable CSV and JSON export capabilities for downstream validation and integration use cases.
- Accelerated test-data availability from days to hours, enabling faster development and validation cycles.
- Tech Stack:* Python, Flask, Metadata Grounding, SQL, JSON/CSV Serialization

PROJECT WORK

GenAI Medical Assistant

June 2025

- Developed an AI assistant using FastAPI and Azure OpenAI for symptom-based reasoning and contextual responses.
- Implemented prompt grounding and retrieval strategies to reduce hallucinations and improve response accuracy.
- Tech Stack:* Python, FastAPI, Azure OpenAI, Angular, Azure Cloud

ShopNest - E-commerce Platform

September 2023

- Designed and developed RESTful microservices using Spring Boot, including authentication and product management modules.
- Implemented event-driven communication using Kafka producers and consumers, enabling asynchronous processing, improving service decoupling, and supporting scalable inter-service workflows.
- Tech Stack:* Java, Spring Boot, Node.js, Angular, MySQL, Kafka, Docker

EDUCATION

National Institute of Technology, Jamshedpur

July 2019 – June 2023

B.Tech in Electrical Engineering

CGPA: 8.12/10

AWARDS AND CERTIFICATES

- Databricks Certified Generative AI Engineer:** Certified in production-grade Generative AI engineering, including Vector Search, Model Serving, MLflow, Unity Catalog, RAG systems, and LLM orchestration.
- Bronze Award (CGI):** Honored for automation initiatives optimizing metadata-driven ETL pipeline deployments and reducing manual operations by 50%.